# Preserving Marker Specificity with Lightweight Channel-Independent Representation Learning

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## Abstract

Multiplex tissue imaging measures dozens of protein markers per cell, yet most deep learning models still apply early channel fusion, assuming shared structure across markers. We present a systematic study of channel fusion as an architectural design choice for supervised and self-supervised representation learning in multiplex imaging. Using a classical Hodgkin lymphoma CODEX dataset with approximately 145,000 cells and 49 markers, we compare early-fusion CNNs with channel-isolated architectures. Our analysis reveals that preserving channel isolation in early layers yields substantially stronger representations than early fusion, particularly for rare-cell discrimination. This advantage is most pronounced under self-supervised pretraining, where early-fusion models show limited ability to retain marker-specific information. Notably, a shallow channel-isolated architecture with only 5.5K parameters exceeds the performance of ResNet variants and matches a 21M-parameter foundation model. We further demonstrate that channel-isolated embeddings enable interpretable phenotyping via marker attribution, producing annotations that respect biological constraints better than conventional pipelines. Our results establish that architectural inductive bias, and specifically delayed channel fusion, can substitute for model scale in multiplex representation learning.
Code is available at `https://github.com/SimonBon/CIM-S`

**Keywords:** Multiplex Imaging, Representation learning, Channel-Isolated Architecture, Phenotyping

## 1. Introduction

Biological multiplex imaging (MI) technologies such as CODEX (Goltsev et al., 2018), MxIF (Gerdes et al., 2013), IMC (Giesen et al., 2014) and MIBI (Angelo et al., 2014) enable the quantification of multiple protein markers at subcellular resolution in biological specimens such as tumor tissue sections. These images provide high-dimensional information on single cell protein marker expression, cell states and spatial tissue organization.

However, due to the complexity of the image data, the interpretation depends on computational methods that can extract marker-specific spatial features and translate them into biologically meaningful phenotypes. Widely used pipelines rely on cell segmentation (Windhager et al., 2023; Bortolomeazzi et al., 2022) followed by aggregation of per-cell marker intensities or handcrafted features. This introduces substantial variability, propagates segmentation errors into downstream analyses (Bruhns et al., 2025; Bai et al., 2021), and limits representations to predefined features rather than learning directly from raw MI data.

Convolutional Neural Networks (CNNs) such as ResNet (He et al., 2016) remain a standard baseline for biomedical image analysis (Xu et al., 2023), including MI. Many CNNs implement early-fusion, in which all input channels are linearly combined in the first convolutional layer and jointly processed thereafter. This inductive bias is well suited to natural images and some multi-channel biomedical modalities. In these settings, channels often encode related measurements and therefore benefit from early joint feature extraction. In single-cell MI, however, each channel corresponds to a molecular marker with distinct biological semantics, and cross-marker relationships are often context-dependent (e.g., cell-phenotype and compartment specific). Consequently, fixed pixel-level mixing in the earliest layers can reduce marker-specific separability and encourage reliance on dataset-specific cross-marker covariance. Despite this mismatch, many CNN adaptations for MI retain early-fusion (Wang et al., 2024).

An alternative line of work, exemplified by NeXtMarker (Gutwein et al., 2025), modifies convolutional architectures to preserve channel-isolated computations in early layers, yielding interpretable feature spaces aligned with marker-specific structure. Although motivated by interpretability, this design is consistent with multimodal learning theory showing that premature fusion can induce modality collapse (Chaudhuri et al., 2025; Wu et al., 2024), where predictive signals from weaker modalities are masked, their gradients diminish, and the model relies on only a subset of inputs. In MI, this would imply that biologically essential markers may contribute little to learning, degrading cell-phenotype discrimination and phenotypic resolution. This risk is smaller in RGB images, where channels are strongly correlated and early-fusion captures shared spatial structure (Dolz et al., 2019), but is more pronounced in single cell MI where channels are weakly correlated at most.

Taken together, these observations suggest that cellular phenotypes depend more on local stoichiometric patterns than on deep hierarchical abstractions. This implies that early-fusion can degrade representation quality in deep learning for MI. We therefore hypothesize that keeping markers independent in the early layers is a key architectural inductive bias for MI. To test this hypothesis, we conducted a comprehensive study of channel fusion as an architectural design choice. We compared deep learning model variants that enforce channel-isolation with early-fusion architectures and a foundational proteomics model (Shaban et al., 2025). We evaluated these approaches in both supervised and self-supervised training setups. All experiments used a publicly available 49-plex CODEX dataset of classical Hodgkin lymphoma (cHL) (Shaban et al., 2024). In addition, we assess whether learned channel-isolated embeddings facilitate interpretable cell phenotyping through Layer-wise Relevance Propagation (Bach et al., 2015).

This work's contributions can be summarized as follows:

1. Systematic analysis of how channel-isolation architectural design affects representational quality, marker retention, and rare-cell discrimination in MI across diverse experimental setups.

2. Comparison of architectural inductive bias versus model scale, testing whether compact architectures can match large pretrained models for MI cell phenotyping.

3. Demonstration of automated phenotyping via propagated channel-wise marker relevance.

## 2. Methods

Our goal is to evaluate whether preserving channel-wise isolation in the early layers of CNNs provides a more suitable inductive bias for single-cell MI representation learning than a conventional early-fusion CNN design. To isolate the effect of fusion strategy, we define a simple baseline architecture that uses channel-isolated processing with optional information fusion. Other architectural and training factors with potential impact on representation quality are held constant or deliberately minimized, such that any performance differences can be attributed primarily to the fusion stage. Additionally, we utilize these channel-isolated architectures to go beyond recovering the cell-phenotype labels provided in existing datasets, and assess whether the learned representations can be leveraged to perform cell-phenotyping for newly generated non annotated datasets.

**Channel-Isolated Model (CIM):** We define a lightweight channel-isolated CNN that preserves channel separation throughout the backbone. Let the input be a $C$-channel multiplex image. The first processing step (stem) is a grouped $3 \times 3$ convolution with $\texttt{groups} = C$, so each input channel is processed in its own group. The stem expands each channel into $k$ feature maps, producing $C \cdot k$ feature maps in total.
The backbone consists of $N$ identical processing blocks. In each block, a grouped $3\times3$ convolution with $\texttt{groups} = C$ processes the $k$ feature maps of each channel group independently, ensuring that no cross-channel information is mixed. We then apply a channel-separated squeeze-and-excitation (SE) module (Hu et al., 2018) that re-weights feature maps within each channel group without coupling different channels. After $N$ blocks, the output tensor is used directly as the learned feature representation. Thus, CIM performs no cross-channel fusion at any stage.

**Channel-Isolated Model with Late Fusion (CIM-LF):** CIM-LF uses the same stem and $N$-block channel-isolated backbone as CIM. The only difference is the output stage. While CIM outputs the backbone representation directly, CIM-LF appends a late-fusion stage that mixes information across channels. Concretely, CIM-LF applies $M$ standard (non-grouped) $3 \times 3$ convolutions, each followed by ReLU activation and max pooling, to fuse features across the $C$ channel groups. Therefore, CIM-LF introduces cross-channel interaction only after the backbone, whereas CIM never introduces cross-channel interaction.

**Employed Configurations:** We use intentionally minimal variants of CIM and CIM-LF, denoted CIM-S and CIM-LF-S. Both share the same core backbone settings ($N = 1$ processing block and $k = 4$ feature maps per input channel), but differ in whether a late-fusion stage is present with $M = 2$ for CIM-LF-S. CIM-S contains ∼5.5K parameters, whereas CIM-LF-S contains ∼698K parameters due to the additional fusion stage. These architectures are kept compact to isolate the effect of the fusion strategy rather than to maximize model capacity. CIM and CIM-LF can be trained either (i) in a supervised setting by attaching a classifier head to the learned feature representation, or (ii) in a self-supervised setting using contrastive methods such as SimCLR (Chen et al., 2020), as illustrated in Fig. 1.

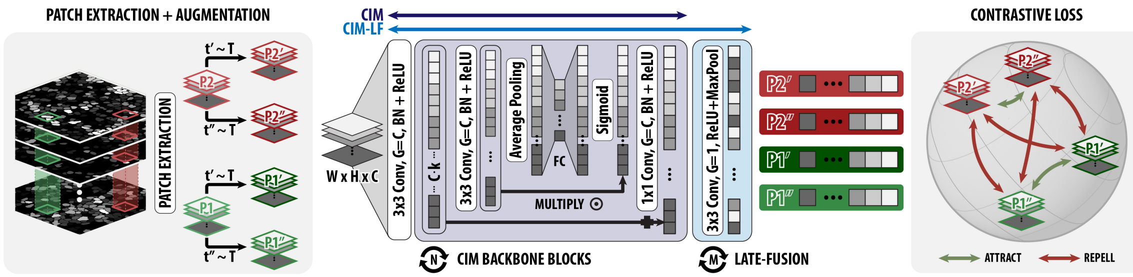


Figure 1: **Left:** Patches ($P1$, $P2$) are extracted from each multiplex image, and augmentations $t', t'' \sim \mathcal{T}$ are sampled to generate two views of each patch. **Center:** CIM and CIM-LF encoders used for feature extraction, consisting of $N$ backbone blocks and (for CIM-LF) $M$ late-fusion blocks. **Right:** Contrastive objective applied to the augmented views of $P1$ and $P2$.

**Cell phenotyping using CIM:** Once the channel-isolated encoders are trained via self-supervised learning, we use the frozen representations to perform label-free cell phenotyping. The process follows three main steps:

1. *Marker attribution via LRP:* To identify which markers drive a cell's latent representation, we apply Layer-wise Relevance Propagation (LRP). Starting from the bottleneck embedding, LRP back-propagates relevance through the encoder to the input channels. We use the EpsilonGammaBox rule to generate marker-specific spatial relevance maps.

2. *Relevance aggregation:* We convert the spatial relevance maps into a single relevance score per marker. We suppress low-magnitude noise and normalize the signal by clipping to the 99$^{\text{th}}$ percentile. The resulting scores are weighted by the raw input intensity to ensure that attribution is grounded in the marker's observed signal.

3. *Phenotype assignment:* We map the marker-wise relevance scores to cell types using a predefined association matrix. For each target phenotype, we aggregate the scores of its characteristic markers (e.g., CD4, CD7 and CD8 for T cells) by averaging the top three marker scores. Each cell is assigned the phenotype with the highest aggregate score, enabling categorization without using ground-truth phenotype labels during training.

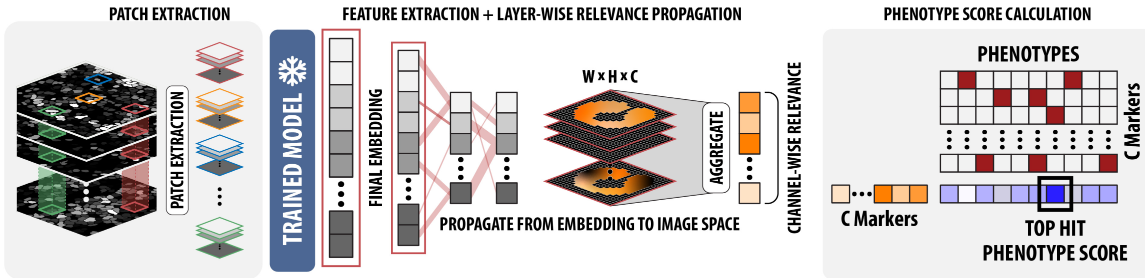


Figure 2: **Left:** Cell-centered patch extraction from multiplex images. **Center:** Pretrained CIM variant encodes each patch into a final embedding; LRP is applied to propagate relevance from the embedding back to image space and aggregated to obtain channel-wise relevance scores. **Right:** Phenotype scoring and assignment by multiplying the channel-wise relevance vector with predefined phenotype marker signatures and selecting the top-scoring phenotype.

## 3. Experiments

We perform two sets of experiments to assess the single-cell representation quality of channel-isolated models: (i) benchmarking CIM models in supervised and self-supervised learning (SSL) settings against early-fusion variants as well as a foundation model for MI, and (ii) generating phenotype labels for an entire dataset using our automated phenotyping pipeline and comparing them to provided annotations derived from conventional segmentation- and expert-driven methods.

### 3.1. Dataset

Throughout this paper, a publicly available CODEX dataset of classical Hodgkin lymphoma (cHL) (Shaban et al., 2024) is used, containing approximately 145,000 cells across 49 protein markers. Shaban et al. provide annotations for 16 phenotypes, generated via integrated marker-expression analysis using DeepCell segmentation (Greenwald et al., 2022), clustering, and expert curation (Fig. S1B–C). Each marker channel is normalized to its 99.9th-percentile intensity.

### 3.2. Experiment 1 – Representation Learning Benchmark

CIM-S and CIM-LF-S are compared against two groups of baselines:

- **Channel-isolated architectures:** NeXtMarker (Gutwein et al., 2025), which preserves channel isolation via grouped convolutions, and, in a separate evaluation, the foundation model KRONOS (Shaban et al., 2025).

- **Early-fusion CNNs:** ResNet18 and ResNet50, chosen due to their widespread use in biomedical imaging (Xu et al., 2023). The first convolutional layer is adapted to accept 49 input channels, while the remainder of each model is kept unchanged.

## Supervised Learning

**Setup:** Each model is trained in a supervised setting by attaching a two-layer fully connected classifier to the feature space and using a 70/20/10 train/validation/test split. During training, patches are augmented using a set of augmentation $\mathcal{T}$: vertical and horizontal flips, affine transformations (rotation and mild scaling), channel-wise intensity scaling, and additive Gaussian noise. A detailed overview of $\mathcal{T}$ can be seen in Tab. S1. Training is performed for 30 epochs using the Adam optimizer (Kingma, 2014) with a learning rate of $10^{-3}$ and weighted cross-entropy to compensate for class imbalance.

**Results:** As shown in Fig. 3 (left), channel-isolated architectures outperform early-fusion CNNs. NeXtMarker achieves with 84.2% the highest balanced accuracy, while the best early-fusion model, ResNet18, reaches 80.8%. We report balanced accuracy to account for rare cell phenotypes (Fig. S1B). While differences in overall accuracy are smaller, balanced accuracy indicates that channel-isolated designs better capture minority phenotypes such as epithelial and mast cells.

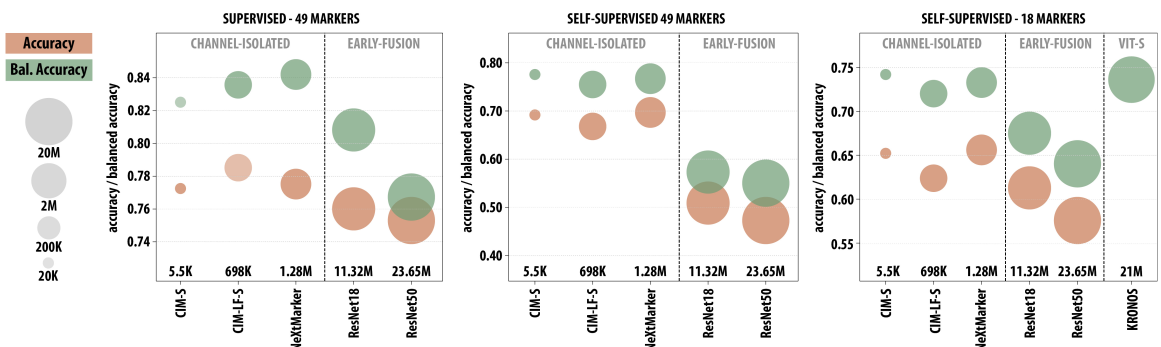


Figure 3: Representation-quality benchmark using accuracy (orange) and balanced accuracy (green). Bubble size indicates parameter count. **Left:** Supervised results using all 49 markers. **Middle:** Self-supervised results using all 49 markers, evaluated by linear probing. **Right:** Self-supervised results using a reduced 18-marker panel, including comparison with the KRONOS foundation model.

## Self-Supervised Learning

**Setup:** For SSL, we adopt the SimCLR (Chen et al., 2020) framework with two augmented views per patch, temperature 0.2, batch size 512, and the LARS optimizer (You et al., 2019). Data augmentations match the supervised setting, with the addition of random channel dropping with probability 0.1 per channel (indicated in Tab. S1; augmentations are kept identical across all models and SSL setups. CIM-S, CIM-LF-S, and NeXtMarker are pretrained for 2,000 iterations, whereas larger early-fusion CNNs are pretrained for 16,000 iterations to account for their higher parameter counts. Under these schedules, models see approximately 4million and 33 million patches, respectively. After pretraining, the encoder is frozen and used as a feature extractor; a two-layer linear classifier is then trained following the supervised protocol. All training procedures utilize a 70%/10%/20% training, validation, test split.

**Results:** The impact of early-fusion vs. channel-isolation is substantially larger in the SSL setting (Fig. 3 (middle); Tab. S2). Early-fusion CNNs trained on all 49 markers reach balanced accuracies of 55–58%, indicating that they learn comparatively weak representations in the absence of labels. In contrast, channel-separable models achieve markedly higher performance: CIM-S reaches 77.6%, slightly exceeding NeXtMarker (76.7%) and CIM-LF-S (75.5%). These results are consistent with the interpretation that early fusion struggles to preserve informative signal across many channels during self-supervised pretraining, whereas channel-isolated processing better retains marker-specific variation that can be exploited by a linear probe.

**Comparison with the foundation model KRONOS:** We compare our approach to the foundation model KRONOS (Shaban et al., 2025), which is evaluated on the same cHL dataset used throughout this study but under a reduced 18-marker protocol. For the 18-marker setting, the SSL protocol is unchanged: all hyperparameters and augmentations are kept identical, and the only difference is the reduced marker set. The qualitative trends observed with the full marker panel are preserved under marker reduction, although performance differences narrow between early-fusion and channel-isolated architectures (Fig. 3 middle and right). Notably early-fusion CNNs improve when the number of markers is reduced, whereas channel-isolated models exhibit the opposite trend. This behavior is consistent with the hypothesis that early fusion underutilizes high-dimensional marker information (e.g., by concentrating on a subset of channels), while channel-isolated processing benefits from preserving marker-specific signal. Under the 18-marker setting, the channel-isolated models match KRONOS performance; CIM-S slightly outperforms KRONOS (74.2% vs. 73.6%) despite a substantially smaller parameter count (CIM-S $\sim$5.5K vs. KRONOS $\sim$21M). Together, these results suggest that an appropriate architectural inductive bias can recover discriminative structure without large parameter budgets or massive pretraining.

### 3.3. Experiment 2: Unsupervised Phenotyping and Label Refinement

Accurate cell-phenotype annotation is a critical step in the analysis of MI data and underpins downstream tasks such as cellular neighborhood characterization and cell–cell interaction analysis. However, phenotype labels in multiplex datasets are often imperfect due to reliance on expert interpretation, segmentation errors, and annotator bias. Here, the goal is to assess whether CIM-S can recover accurate phenotypes directly from raw multiplex images. To this end, the CODEX cHL dataset is treated as a newly acquired, unlabeled cohort. The phenotyping pipeline described in Section 2 is applied using LRP-based channel-wise relevance aggregation (channel-wise spatial LRP examples shown in Fig. S3), and the resulting phenotype labels are compared to the dataset's provided phenotype labels, which were generated using DeepCell (Greenwald et al., 2022) segmentation, clustering, and expert curation (Shaban et al., 2024).

**Setup:** We used the self-supervised CIM-S backbone from the 49 marker setup as a fixed feature extractor and as the basis for the downstream phenotyping pipeline. Cell locations for patch extraction were obtained with Cellpose (Stringer et al., 2021) by extracting the cell-probability map yielded by Cellpose and identifying local maxima corresponding to cell centers. We then extracted $20 \times 20$ pixel patches centered at these coordinates and treated each patch as a single-cell instance. Using this procedure, we obtained approximately 101K single-cell patches. Phenotype definitions (Fig. S4) were specified by an immunology expert and aligned with established guidelines and recent literature (Maecker et al., 2012; Cossarizza et al., 2019; Watanabe et al., 2023; Kaminski et al., 2012). All inference was performed on a single *NVIDIA* V100 32GB GPU.

**Results:** Cell phenotyping on the full cHL slide using our automated pipeline took approximately 3 minutes, consistent with the lightweight design of CIM-S (5.5K parameters) and its minimal memory footprint. A qualitative overview of the assigned cell phenotypes is shown in Fig. 4 and Fig. S1A. In the left panel, a tumor–immune border region is depicted: CD20 aligns with B cells, CD4 with T cells, and CD30 with cHL tumor cells, as expected. In the right panel, B-cell and T-cell marker signal again co-localizes with the corresponding cell-type annotations. In this region, endothelial phenotype assignments also correspond closely to CD31 signal, consistent with CD31 as an endothelial marker. We note that the segmentation masks are included for visualization only and are not used during model training or inference.

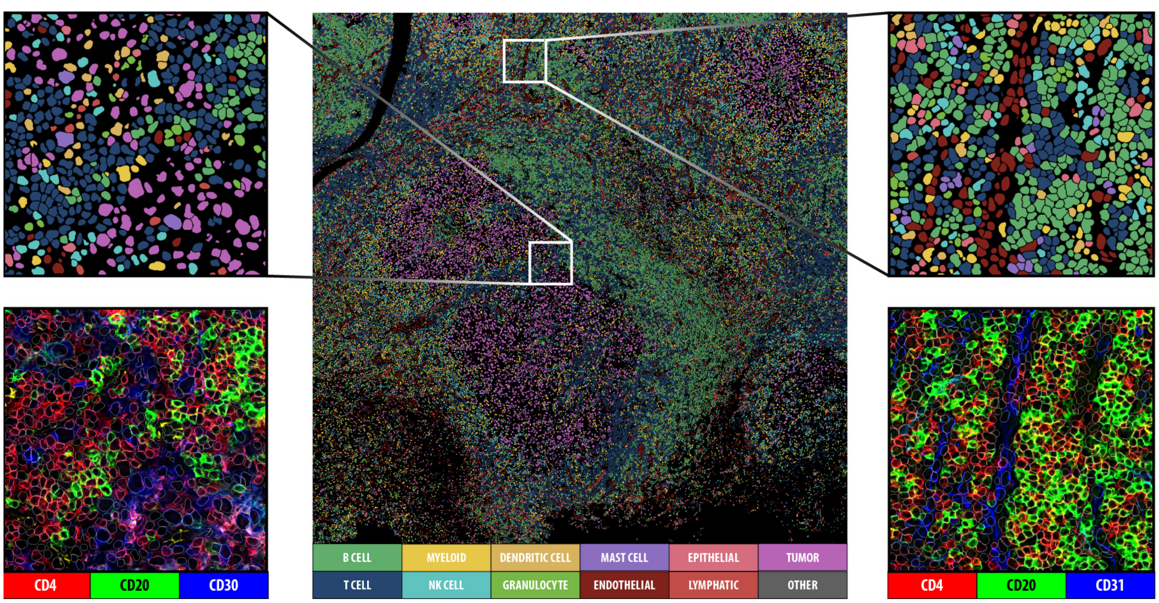


Figure 4: Qualitative assessment of cell-type annotations on the cHL CODEX dataset, showing detailed views of a tumor–immune border region and a region traversed by a vessel. Segmentation masks are utilized for visualization purposes only.

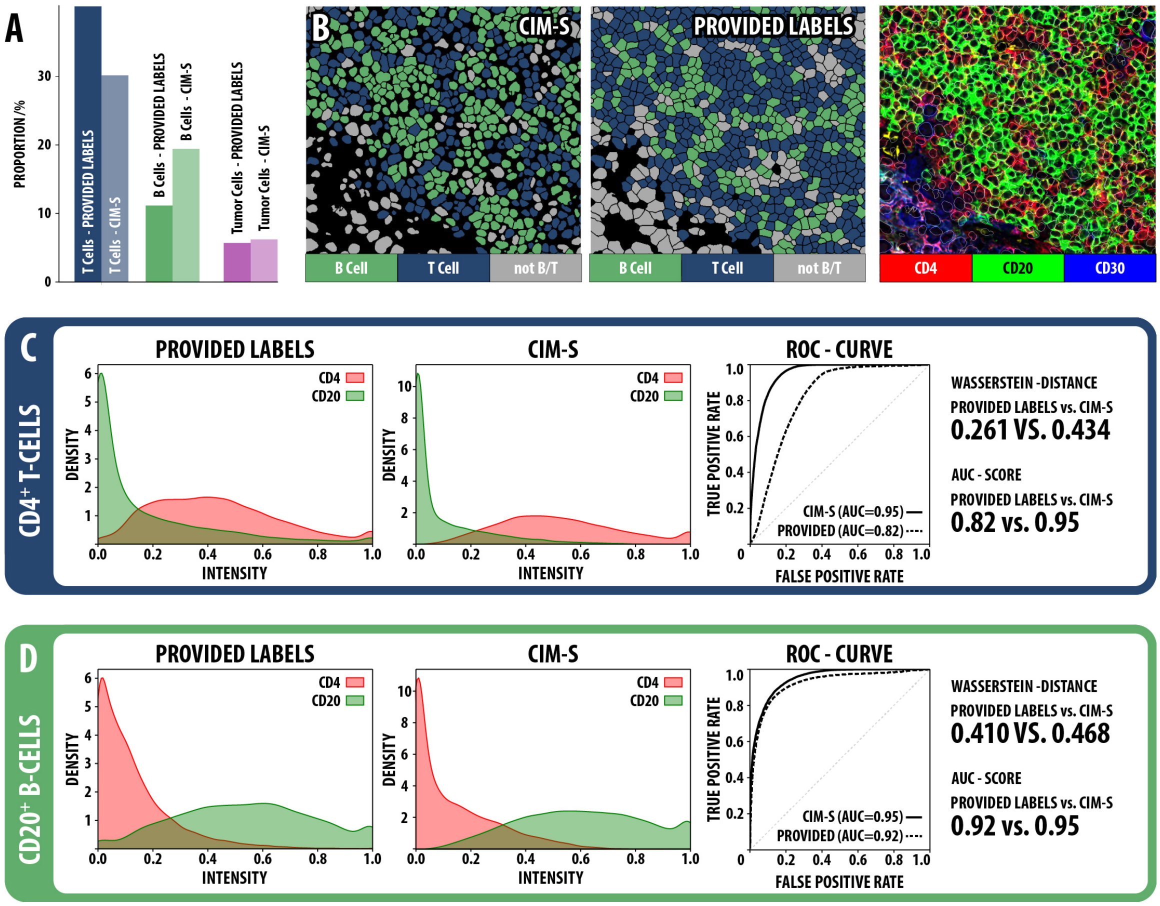


Figure 5: Quantitative comparison between labels created with CIM-S and the provided labels on the cHL dataset. **A:** Differences in inferred cell-type proportions. **B:** Representative region illustrating the discrepancy. **C/D:** CD4/CD20 separability within the $CD4^+$ T-cell and B-cell populations defined by each label set, quantified by Wasserstein distance and ROC/AUC.

**Evaluating label reliability:** Comparing labels created with CIM-S + LRP to the provided labels revealed a strong shift in immune cell annotations: CIM-S + LRP identified 72.7% more B cells and 24.8% fewer T cells than the provided labels (Fig.5A), with a representative example shown in Fig.5B. To assess whether this discrepancy reflects model error or limitations in the provided labels, we leveraged a fundamental biological constraint: CD4 (T-cell marker) and CD20 (B-cell marker) are mutually exclusive and should therefore not be co-expressed on the same cell. Accordingly, a correctly phenotyped T-cell population should exhibit high CD4 and low CD20, whereas a correctly phenotyped B-cell population should exhibit high CD20 and low CD4.

We therefore quantified how well CD4 and CD20 signals separate within the T-cell and B-cell populations defined by labels created with CIM-S, and compared this to the corresponding populations defined by the provided labels. Separability was assessed using the Wasserstein

distance (WD) and ROC/AUC analysis for T and B cells respectively (Fig.5C/D). For the T-cell population, CIM-S achieved higher CD4/CD20 separation than the provided labels (WD: 0.434 vs. 0.261), suggesting that a subset of CD20-positive B cells may be misclassified as T cells in the provided labels. For the B-cell population, separability remained high under CIM-S (WD: 0.468) and was comparable to the provided labels (WD: 0.410), indicating that improved T-cell specificity did not compromise B-cell identification. This trend is mirrored by ROC/AUC results: for T cells, CIM-S achieves an AUC of 0.95 versus 0.82 for the provided labels; for B cells, CIM-S reaches 0.95 versus 0.92.
Collectively, these results suggest that the channel-independent architecture acts as an effective denoising filter, producing phenotype assignments that better respect CD4/CD20 exclusivity than assignments derived directly from raw intensity and handcrafted features using cell segmentation.

**Blind expert validation:** To adjudicate between the conflicting assignments, we performed an additional blind evaluation with an external immunology expert. We selected image crops containing disagreements between labels assigned using CIM-S and the provided labels, comprising approximately 100 cells. The expert manually annotated these cells using only the signal for T-cell and B-cell markers, without access to either the provided labels or the CIM-S predictions. Agreement with the provided labels was low (accuracy: 54.1%), indicating systematic errors in the provided labels. In contrast, agreement with labels created with CIM-S was substantially higher (accuracy: 82.1%). This independent assessment supports that the discrepancies introduced by CIM-S largely reflect corrections of noisy provided labels.

## 4. Discussion

**Preserving channel isolation improves multiplex representations:** Our results support the hypothesis that preserving channel isolation in early feature extraction primarily drives representation quality in MI. Across supervised and self-supervised settings, channel-separable networks consistently yield higher balanced accuracy than early-fusion CNNs, indicating improved recovery of minority phenotypes. This pattern aligns with multimodal representation collapse: when heterogeneous inputs are fused early, weaker modalities can become masked and contribute diminishing gradients during training (Chaudhuri et al., 2025; Wu et al., 2024). In MI, where channels reflect distinct molecular measurements with weak inter-channel correlations and marker-specific noise, early mixing risks entangling informative and confounded signals as marker dimensionality increases. Because cellular phenotypes are determined by local combinations and relative intensities of specific markers, architectures that maintain these stoichiometric patterns for longer–such as CIM-S, CIM-LF-S and NeXtMarker–appear better suited for this domain. Our results provide empirical evidence that multiplex-specific architectural priors can preserve marker-level signal and improve rare-cell discrimination under realistic panel sizes.

**Efficient architectures and benchmarking practice:** The CIM-S, CIM-LF-S and NeXtMarker results show that the benefits of marker-aware inductive bias do not depend on large model capacity. Under the KRONOS 18-marker benchmark, shallow backbones achieve balanced accuracy comparable to a foundation-scale transformer while using orders of magnitude fewer parameters. This underscores one of the main takeaways of this study: in MI, preserving channel-isolation can substitute for scale. Our phenotyping analysis also highlights why aggregate classification scores alone may be insufficient: shifts in inferred B- and T-cell proportions and relevance–intensity discrepancies for CD4 and CD20 suggest that high performance against a single reference labeling can still mask biologically ambiguous or artifact-influenced categories. This aligns with the "gold standard paradox" in digital pathology (Aeffner et al., 2017; Rumberger et al., 2024). We therefore recommend that future multiplex benchmarks complement supervised metrics with uncertainty-aware evaluation and attribution plausibility checks.

**Implications for cell-phenotyping:** Our cell-phenotyping experiment indicates that channel-isolated embeddings, combined with LRP, enable cell-level labels directly from multiplex images. CIM-S yields spatially coherent phenotype maps that follow expected marker–cell type relationships and produces marker-relevance profiles that provide interpretable support for its assignments. In cases where our relevance-based annotations diverge from the provided labels, our analyses suggest that these discrepancies are plausible instances of phenotyping error, potentially attributable to segmentation artifacts. Collectively, these results suggest that interpretable, channel-isolated models can complement segmentation- and intensity-driven workflows and offer a practical route to phenotype discovery when reference labels are unavailable or unreliable.

**Limitations and future directions:** This study demonstrates that early channel isolation consistently improves multiplex representation learning across both supervised and self-supervised paradigms. While our current evidence is focused on a single cHL CODEX dataset, we have ensured internal robustness by evaluating the framework across different marker densities (18 vs. 49) and learning setups. Validating these findings across diverse tissues and imaging platforms, which vary in spatial resolution and signal-to-noise ratios, remains a key priority for future work. Furthermore, while our relevance-guided phenotyping shows strong alignment with expert annotations, larger-scale and orthogonal validation is needed to support specific clinical applications. Future research will explore hybrid channel-isolated models under large-scale pre-training. Given the computational efficiency of CIM-S, such extensions are highly practical and provide a clear path toward establishing systematic robustness across different batches and acquisition sites.

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| Augmentation | Parameters | Applies to |
|---|---|---|
| Random flip | $p = 0.5$ (horizontal) and $p = 0.5$ (vertical) | Supervised; SSL |
| Random affine | $\theta \sim \mathcal{U}[0, 360]^\circ$; $s \sim \mathcal{U}[0.8, 1.2]$ | Supervised; SSL |
| Intensity scaling | $a \sim \mathcal{U}[0.9, 1.2]$ | Supervised; SSL |
| Gaussian noise | $\mu = 0$; $\sigma \sim \mathcal{U}[0, 0.02]$ | Supervised; SSL |
| Channel drop | drop_p = 0.1 | SSL |

Table S1: Data augmentation parameters. $\mathcal{U}[a, b]$ denotes a uniform distribution on $[a, b]$; parameters are sampled independently per view. Patches are clipped to [0, 1] after augmentation.

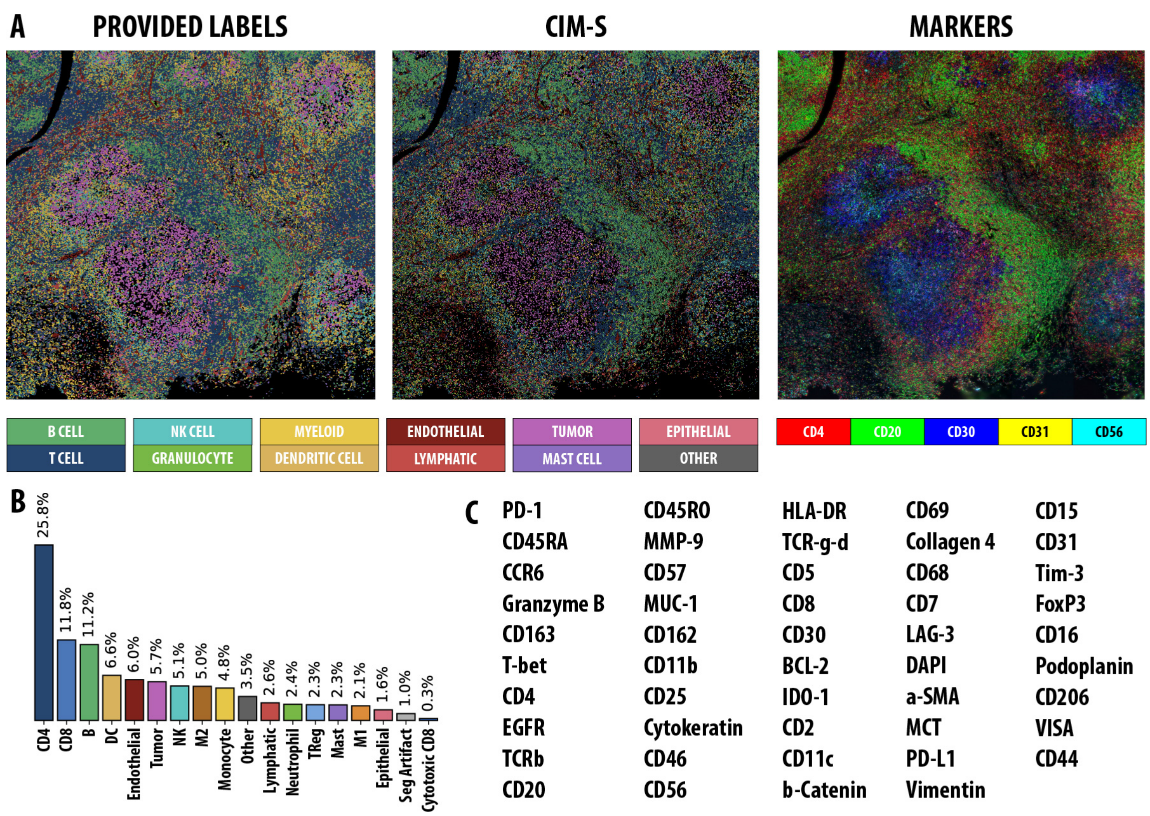


Figure S1: **A**: Cell segmentations colored by the reference labels, our CIM-S annotations, and the corresponding raw marker expression for CD4, CD20, CD30, CD31, and CD56. **B**: Distribution of annotated cell types (colors match panel A). **C**: Complete 49-marker panel used in the dataset.

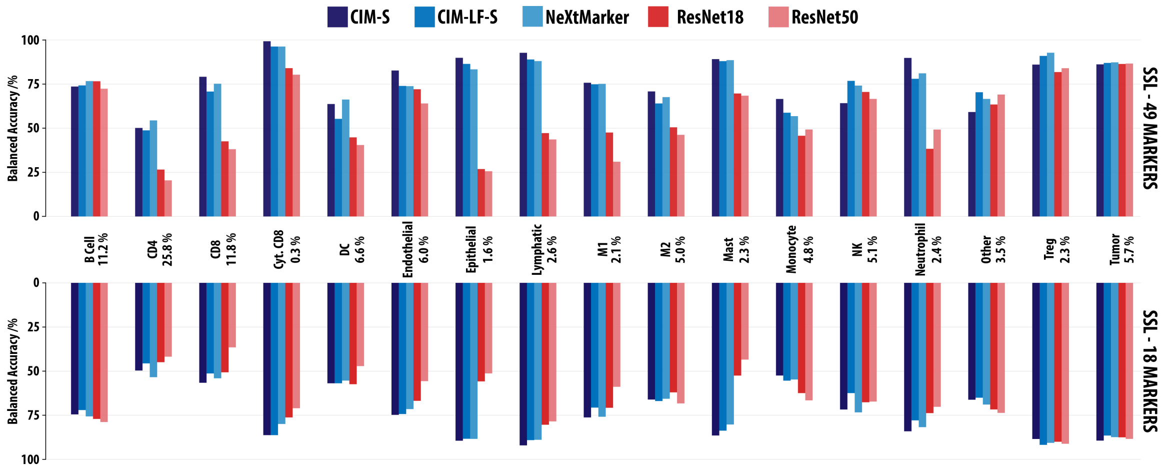


Figure S2: Per-cell-type accuracy for CIM-S, CIM-LF-S, NeXtMarker and ResNet-18/50 under the self-supervised learning setup. The top and bottom panels report results for the 49-marker and 18-marker panel configurations, respectively.

| **Model** | **Supervised** | | **SSL 49 markers** | | **SSL 18 markers** | |
|---|---|---|---|---|---|---|
| | **Acc** | **Bal. Acc** | **Acc** | **Bal. Acc** | **Acc** | **Bal. Acc** |
| CIM-S | 77.3 | 82.5 | 69.2 | **77.6** | 65.2 | **74.2** |
| CIM-LF-S | **78.5** | 83.6 | 66.8 | 75.5 | 62.4 | 72.0 |
| NeXtMarker (Gutwein et al., 2025) | 77.5 | **84.2** | **69.7** | 76.7 | **65.6** | 73.3 |
| KRONOS (Shaban et al., 2025) | – | – | – | – | – | 73.6 |
| ResNet18 | 76.0 | 80.8 | 50.9 | 57.3 | 62.3 | 67.5 |
| ResNet50 | 75.3 | 76.7 | 47.3 | 55.0 | 57.5 | 64.0 |

Table S2: Performance comparison across architectures and training setups (all values in %). Best score per column in bold.

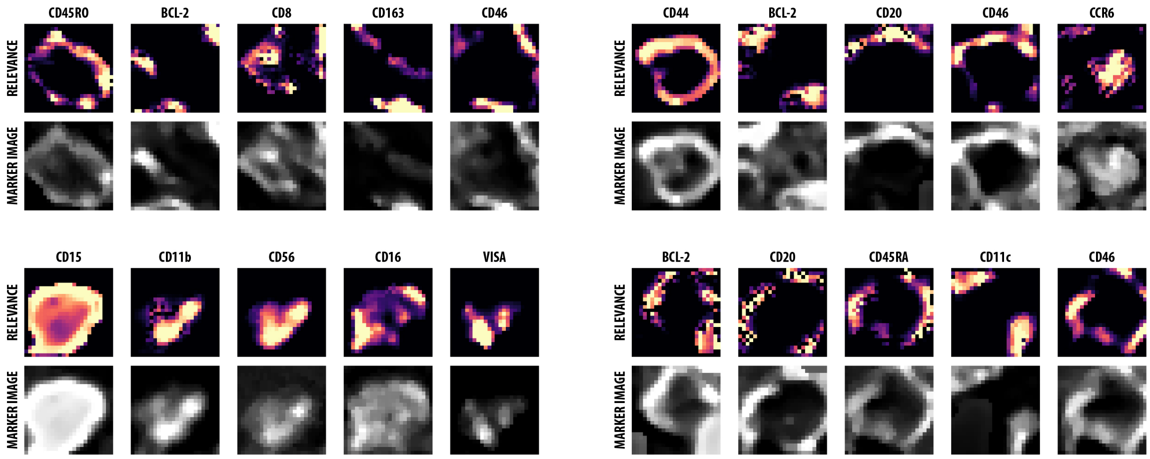


Figure S3: Representative LRP visualizations for CIM-S. For each example, the five channels with the strongest relevance are shown alongside the corresponding marker intensity image.

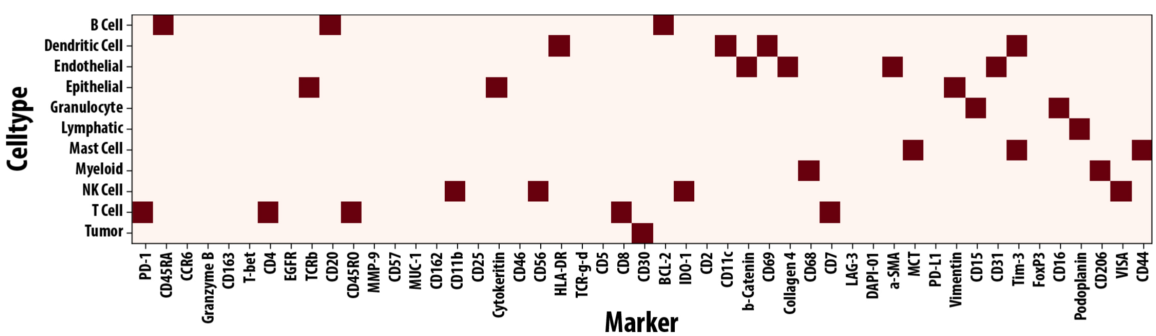


Figure S4: Module definitions used for phenotyping. Red squares denote markers included in each cell phenotype module; all other markers are ignored for that module.